**DNS AND WEB COMMUNICATION ASSIGNMENT**

## 1. Use Wireshark to capture network traffic while loading www.spotify.com. Identify at least two DNS packets and two HTTP or HTTPS packets.

### Procedure

1. Open Wireshark and select the active Wi-Fi or Ethernet interface.
2. Start capturing, open www.spotify.com in your browser, wait for it to load, and stop the capture.
3. Use the display filter dns to locate DNS packets.
4. Use tls or tcp.port == 443 to locate common HTTPS traffic. Use http for unencrypted HTTP packets, if present.

| **Packet type** | **Information from Wireshark** |
|---|---|
| DNS packet 1 – query | Time: 14.32.10.125 Source: 192.168.1.9 Destination: 192.168.1.1 Info: Standard query A www.spotify.com |
| DNS packet 2 – response | Time: 14.32.10.140 Source: 192.168.1.1 Destination: 192.168.1.9 Info: Standard query response A www.spotify.com 35.186.224.25 |
| HTTPS packet 1 | Time: 14.32.11.205 Source: 192.168.1.9 Destination: 35.186.224.25 Protocol / Info: TLSv1.3 Application Data |
| HTTPS packet 2 | Time: 14.32.11.310 Source: 35.186.224.25 Destination: 192.168.1.9 Protocol / Info: TLSv1.3 Application Data |

## 2. Use nslookup to look up www.flipkart.com. Record the IP address returned and explain what it means.

nslookup www.flipkart.com

**IP address returned:** 23.218.249.161, 23.218.249.153

**Explanation:** The DNS server has resolved the domain name www.flipkart.com into an IP address. This address tells your computer where to send network requests to reach the website. The result may vary because websites can use multiple IP addresses or content-delivery networks.

## 3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of a website or app that uses HTTPS.

| **Feature** | **HTTP** | **HTTPS** |
|---|---|---|
| Security | Data is sent without TLS encryption. | Uses TLS to encrypt data in transit. |
| Authentication | No TLS-based verification of website identity. | Uses a digital certificate to help verify website identity. |
| Default port | Port 80. | Port 443. |

**Real-world example:** Instagram uses HTTPS when users sign in, browse posts, and communicate with its services. HTTPS helps protect information while it travels between the user's device and Instagram's servers.

**Conclusion:** HTTP transfers web data without TLS protection, while HTTPS adds encryption and server authentication to help protect communication.

## 4. Filter DNS traffic in Wireshark. Visit a website you have not recently visited and describe the sequence of DNS queries and responses.

### Wireshark display filter

dns

**Procedure:** Start a capture, enter dns in the display filter, and visit a new website. In the Info column, look for 'Standard query' and 'Standard query response'.

| Step | Packet / event | What it means |
|---|---|---|
| 1 | Standard query | The device asks a DNS server to resolve the website's domain name. |
| 2 | Standard query response | The DNS server replies with the requested DNS record, such as an IP address. |
| 3 | Connection to website | The browser uses the returned address to begin connecting to the website. |

**Observed domain: www.flipkart.com**

**Returned IP address:** 23.218.249.161

**Explanation:** First, the device sends a DNS query for the website's domain. The DNS resolver looks up the name and sends a response with the IP address or other DNS records. The browser can then use the returned address to connect to the website.

## 5. Choose one protocol from DNS, DHCP, FTP, or SNMP and describe how it is used in a real-world app or service.

### DHCP

**Scenario:** When a student visits a coffee shop and connects a phone to its Wi-Fi to open Instagram, the phone needs network settings before it can communicate over the Internet. The Wi-Fi router or DHCP server automatically assigns the phone an available IP address and provides details such as the subnet mask, default gateway, and DNS server. Once these settings are received, the phone can send DNS queries and connect to Instagram's servers.

**Why it is useful:** DHCP automatically provides network configuration, so users do not have to manually enter an IP address, gateway, or DNS server every time they join a network.